

Zavier Vega-Yu

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EDUCATION

Texas A&M University

Bachelor of Science in Computer Science

Expected May 2027

College Station, TX

- **GPA:** 3.97/4.00 | 4x Dean's Honor Roll
- **Coursework:** Operating Systems, Distributed Systems, Computer Networks, Computer Architecture, Database Systems, Data Structures, Algorithms, Machine Learning, Artificial Intelligence, Information Storage & Retrieval

EXPERIENCE

Oracle Cloud Infrastructure

Software Engineer Intern | Applied AI Solutions

Jun. 2026 – Sep. 2026

Austin, TX

- Built core backend capabilities for an **agentic platform** that generates cloud architectures and **Terraform** plans
- Designed an OCI **Terraform** cost-estimation package spanning TypeScript, a Go API, and a React UI for FinOps
- Implemented **idempotent recovery** across **agent planning stages** with durable state and safe retries

Texas A&M University

Undergraduate Teaching Assistant | Department of Computer Science

Aug. 2025 – May 2026

College Station, TX

- Automated grading with **10+** autograders in **Python, Bash, Docker**, enforcing correctness and time limits
- Led weekly labs and office hours for **100+** students covering **C++**, memory management, and data structures

Hewlett Packard Enterprise

Software Engineer Intern | HPE Nonstop Solutions

May 2025 – Aug. 2025

Fort Collins, CO

- Parallelized HPE Nonstop software and firmware updates, reducing benchmarked maintenance time **30%**
- Engineered a **contention-aware task scheduler** in **C++**, increasing firmware update throughput by **10%**
- Eliminated identified race/deadlock paths via **thread-safe synchronization** and concurrency stress tests

LEADERSHIP

Tidal TAMU | President | Computer Science Student Organization

Aug. 2023 – Present

- Direct **30+ officers** across technical and operations teams; Named **2025 Student Organization of the Year**
- Organize MLH-affiliated hackathons averaging **300+ participants per event** from universities across the U.S.
- Engineered a full-stack event management platform serving **2,000+ users** with **React, TypeScript, Node.js, and PostgreSQL**, supporting registration and admin operations

PROJECTS

MLLM-IPI | Python, PyTorch, Hugging Face, Qwen-VL, SAM, Streamlit

Jun. 2026

- Built an AI security pipeline evaluating **image-based prompt injection** against Qwen2.5-VL and Qwen3-VL
- Automated experiment sweeps measuring **attack success rate, MSE, and SSIM** across rendering strategies
- Integrated **SAM** segmentation to rank regions and adaptively place multimodal prompts for robustness testing

CreditMob | TypeScript, Python, React, Firebase, Selenium

- Implemented a credit platform with personalized card recommendations and AI-guided financial education
- Ranked cards by user fit using **vector embeddings** and **cosine similarity** across profile attributes
- Published an **open-source npm package** with a Selenium pipeline for normalized credit-card data and API access

Multithreaded Web Crawler | C++, WinSock, TCP/IP, HTTP

- Built a multithreaded C++ web crawler processing **10,000+ URLs** using low-level HTTP networking
- Coordinated concurrent workers over a shared URL frontier with thread-safe synchronization and link extraction
- Implemented **robots.txt** parsing and enforcement to respect site-specific crawling policies

TECHNICAL SKILLS

Languages: C, C++, Java, Python, Go, Rust, TypeScript, JavaScript, SQL, x86 Assembly

Frameworks: React, Spring Boot, Node.js, FastAPI, gRPC, REST, POSIX, PostgreSQL, Redis, Kafka, Prisma

Infrastructure: Linux, Git, AWS, OCI, GCP, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD

AI/ML: Codex, Claude Code, Cursor, LLM APIs, MCP, RAG, Langchain, TensorFlow, PyTorch, Pinecone